

Air-Gap & Disconnected Migration

VMware to KVM in Secure Environments Without Internet

Classified networks, SCIF facilities, OT/ICS environments, and air-gapped data centers face unique migration challenges. hyper2kvm provides a fully offline migration workflow with encrypted media transfer, chain-of-custody tracking, and FIPS-validated operations.

SCIF | OT/ICS | FIPS 140-2 | Sneakernet | Offline Repos | Two-Person Integrity

Air-Gap Challenge Overview

Why disconnected environments exist and why VMware licensing makes migration urgent.

Disconnected Environment Types

Environment	Classification	Network Policy	VMware Pain Point	Migration Challenge
SCIF / SCI	TS/SCI	Complete air-gap, no internet ever	License activation requires phone-home; \$240K+/yr for 50 VMs	No package repos, no driver downloads, no online activation
DoD / IL4-6	CUI to Secret	NIPR/SIPR isolated segments	Broadcom audit compliance across classification boundaries	Cross-domain transfer requires data diode or manual review
OT / ICS / SCADA	Industrial	Purdue model L3.5 DMZ	ESXi patches delayed 6-12 months for stability testing	24/7 uptime requirement, no maintenance windows > 4 hours
Healthcare / HIPAA	PHI	Network segmentation, no cloud	Per-VM licensing doubles cost for HA pairs	HIPAA audit trail must be maintained during migration
Financial / PCI-DSS	PCI Zone 1	Cardholder data environment isolated	Annual PCI re-certification with new Broadcom terms	Change control windows restricted to 2-hour slots
Nuclear / NRC	Safeguards	Complete air-gap, hardware tokens	10CFR73.54 cyber security plan requires approved software list	Every binary must be on approved software list before install

VMware Air-Gap Cost Problem

Current VMware Air-Gap Costs (50 VMs)

- vSphere Enterprise Plus: \$6,175/CPU x 8 CPUs = \$49,400/yr
- vCenter Server: \$8,975/instance + \$2,250 SnS
- vSAN: \$5,575/CPU x 8 CPUs = \$44,600/yr
- NSX: \$6,995/CPU x 8 CPUs = \$55,960/yr
- Offline license activation: manual key + phone support
- Air-gap patch media: \$15,000/yr custom ISO delivery

KVM Air-Gap Costs (Same 50 VMs)

- RHEL KVM (or AlmaLinux/Rocky): \$0-\$13,000/yr
- libvirt + QEMU: \$0 (included in OS)
- No license activation needed (open source)
- No phone-home, no telemetry, no audit surprises
- Offline repos: createrepo from ISO media
- hyper2kvm migration: \$0 (Apache-2.0 license)

- Total: \$176,185/yr for 50 VMs

- Total: \$0-\$13,000/yr — savings up to \$163K/yr

Offline Media Transfer Workflow

USB/HDD sneakernet with encrypted disk images, checksums, and chain-of-custody.

End-to-End Sneakernet Process

```
# STEP 1: Export VMs on source network (VMware side)
h2kvmctl export --source vsphere --server vcenter.classified.mil \
  --vm "db-server,app-server,web-frontend" \
  --output /mnt/transfer-media/ \
  --encrypt --passphrase-file /root/transfer-key.gpg \
  --checksum sha256 --manifest

# STEP 2: Verify checksums before physical transfer
h2kvmctl verify --manifest /mnt/transfer-media/manifest.json
# Output: 3 VMs verified, 47.2 GB total, SHA-256 checksums match

# STEP 3: Physical transfer (write-blocker → encrypted USB → courier)
# Chain-of-custody form signed by: exporter, courier, importer

# STEP 4: Import on destination network (KVM side)
h2kvmctl import --source /mnt/transfer-media/ \
  --decrypt --passphrase-file /root/transfer-key.gpg \
  --verify-checksums \
  --output /var/lib/libvirt/images/ \
  --convert qcow2 --compress \
  --emit-domain-xml --deploy-libvirt
```

Transfer Media Options

Encrypted USB 3.2 SSD (Recommended)

- Hardware: IronKey D500S or Apricorn Aegis Padlock
- Capacity: 1-8 TB, hardware AES-256 encryption
- Transfer speed: 1 GB/s (USB 3.2 Gen 2)
- FIPS 140-3 Level 3 certified (IronKey)
- Tamper-evident: epoxy-filled, brute-force self-destruct
- 50 VMs at 50 GB avg = 2.5 TB → 1x 4TB drive, ~42 min transfer

Data Diode Transfer (Highest Security)

- Hardware: Owl Cyber Defense DualDiode or Waterfall Security
- One-way transfer: physically impossible to exfiltrate
- Throughput: 100 Mbps - 10 Gbps depending on model
- h2kvmctl streams VM data through diode protocol
- Receiving side reassembles and verifies checksums
- Required for: cross-domain (e.g., UNCLASS → SECRET)

Manifest & Chain of Custody

Encryption Options

- manifest.json: VM name, size, SHA-256, timestamp, exporter ID
- GPG-signed manifest prevents tampering during transport
- DD Form 1149 or equivalent for physical media tracking
- Two-person integrity (TPI) for classified media handling
- Media sanitization after transfer: DoD 5220.22-M or degauss
- Audit log: who exported, who transported, who imported

- LUKS2 full-disk: `cryptsetup luksFormat /dev/sdX`
- GPG per-file: `gpg --symmetric --cipher-algo AES256`
- `h2kvmctl --encrypt`: uses AES-256-GCM with PBKDF2
- FIPS mode: `fips-mode-setup --enable` on RHEL/Rocky
- Key management: HSM or split-knowledge passphrase
- At-rest + in-transit encryption for complete chain

RPM/DEB Package Bundling

Pre-staging all dependencies for offline installation on target hosts.

Building Offline RPM Repository

```
# On internet-connected staging machine (same OS version as target)
# Download all KVM/libvirt packages and dependencies
dnf download --resolve --destdir=/mnt/offline-repo/ \
    qemu-kvm libvirt libvirt-client virt-install virt-viewer \
    libguestfs-tools python3-libvirt bridge-utils \
    edk2-ovmf swtpm swtpm-tools

# Download hyper2kvm and its Python dependencies
pip download -d /mnt/offline-repo/python/ \
    pyyaml lxml requests tqdm paramiko

# Create repository metadata
createrepo_c /mnt/offline-repo/

# Copy to transfer media
cp -r /mnt/offline-repo/ /mnt/transfer-media/repo/
```

Target Host Installation (Offline)

```
# On air-gapped target host
# Mount transfer media
mount /dev/sdX1 /mnt/transfer-media

# Add offline repo
cat > /etc/yum.repos.d/offline-kvm.repo <<'EOF'
[offline-kvm]
name=Offline KVM Repository
baseurl=file:///mnt/transfer-media/repo
enabled=1
gpgcheck=0
EOF

# Install everything offline
dnf install --disablerepo='*' --enablerepo=offline-kvm \
    qemu-kvm libvirt libvirt-client virt-install

# Install hyper2kvm
pip install --no-index --find-links=/mnt/transfer-media/repo/python/ hyper2kvm
```

```
# Verify installation
h2kvmctl doctor --offline
```

Package Bundle Sizes

Component	RHEL 9 / Rocky 9	Ubuntu 22.04	Dependencies	Notes
qemu-kvm	18 MB	22 MB	~45 packages	Core hypervisor
libvirt stack	12 MB	15 MB	~30 packages	VM management daemon
VirtIO drivers (Windows)	580 MB	580 MB	ISO image	virtio-win.iso from Fedora
EDK2/OVMF (UEFI)	4 MB	5 MB	2 packages	UEFI firmware for VMs
hyper2kvm + Python deps	8 MB	8 MB	~15 packages	Migration toolkit
swtpm (vTPM)	2 MB	2 MB	~5 packages	For Windows 11 / BitLocker VMs
Total offline bundle	~650 MB		~100 packages	Fits on any USB stick

Doctor Offline Mode

h2kvmctl doctor --offline verifies all required packages are installed, checks QEMU/KVM kernel module, validates libvirt socket, tests VirtIO driver availability, and confirms UEFI firmware — all without any network access. Reports pass/fail for 15 checks with remediation instructions referencing the offline repo.

Container Image Pre-Staging for KubeVirt

Air-gapped Kubernetes clusters need container images pre-loaded before KubeVirt deployment.

Air-Gapped Container Registry Setup

```
# On internet-connected machine: pull all KubeVirt images
KUBEVIRT_VERSION=v1.3.0
IMAGES=(
  "quay.io/kubevirt/virt-operator:${KUBEVIRT_VERSION}"
  "quay.io/kubevirt/virt-api:${KUBEVIRT_VERSION}"
  "quay.io/kubevirt/virt-controller:${KUBEVIRT_VERSION}"
  "quay.io/kubevirt/virt-handler:${KUBEVIRT_VERSION}"
  "quay.io/kubevirt/virt-launcher:${KUBEVIRT_VERSION}"
  "quay.io/kubevirt/cdi-operator:${KUBEVIRT_VERSION}"
  "quay.io/kubevirt/cdi-controller:${KUBEVIRT_VERSION}"
  "quay.io/kubevirt/cdi-importer:${KUBEVIRT_VERSION}"
  "quay.io/kubevirt/cdi-uploadproxy:${KUBEVIRT_VERSION}"
)

# Save all images to a tar archive
for img in "${IMAGES[@]"; do
  skopeo copy "docker://${img}" "oci-archive:/mnt/transfer/kubevirt-images.tar:${img##*/}"
done
```

Loading into Air-Gapped Registry

Harbor Registry (Recommended)

- Deploy Harbor offline installer (500 MB bundle)
- Self-signed TLS certs (no external CA needed)
- LDAP/AD integration for auth (air-gapped AD)
- Vulnerability scanning with offline CVE DB
- Load images: `skopeo copy oci-archive:kubevirt.tar docker://harbor.local/kubevirt/virt-operator:v1.3.0`
- Image replication between air-gapped sites

K3s Embedded Registry

- K3s supports pre-loading from `/var/lib/rancher/k3s/agent/images/`
- Place tar files: `cp kubevirt-images.tar /var/lib/rancher/k3s/agent/images/`
- K3s auto-imports on restart (no registry needed)
- Simplest option for single-node or small clusters
- No registry infrastructure to maintain
- Add `--private-registry` for multi-node with local registry

CDI Offline Import for VM Disks

- CDI (Containerized Data Importer) needs local source

Image Bundle Sizes

- KubeVirt operator stack: ~2.1 GB (9 images)

- Upload via `virtctl image-upload` from local file
- `virtctl image-upload pvc vm-disk --size=50Gi --image-path=disk.qcow2`
- Or use `DataVolume` with registry source pointing to Harbor
- Pre-convert with `h2kvmctl`, then upload converted QCOW2
- Supports QCOW2, raw, VMDK formats

- K3s binary + images: ~450 MB
- Harbor offline installer: ~500 MB
- Longhorn (storage): ~800 MB
- MetalLB + Multus (networking): ~200 MB
- **Total K8s+KubeVirt bundle: ~4 GB**

Conversion in Disconnected Mode

h2kvmctl performs all conversion operations locally without any network access.

Offline Conversion Pipeline

```
# Full offline conversion – no network calls at any stage
h2kvmctl convert \
  --source /mnt/transfer-media/exported-vms/db-server.vmdk \
  --output /var/lib/libvirt/images/ \
  --format qcow2 \
  --compress \
  --offline \
  --virtio-drivers /mnt/transfer-media/drivers/virtio-win.iso \
  --fix-fstab \
  --fix-initramfs \
  --emit-domain-xml \
  --deploy-libvirt

# What --offline does:
# 1. Disables all network checks (DNS, NTP, package repos)
# 2. Uses local VirtIO drivers instead of downloading
# 3. Rebuilds initramfs using only installed kernel modules
# 4. Skips online activation checks for Windows VMs
# 5. Uses local timezone data instead of NTP sync
```

Offline Capabilities by Source Format

Source Format	Conversion	VirtIO Inject	initramfs Fix	fstab Fix	Domain XML	Boot Test
VMDK (VMware)	Full	Yes	Yes	Yes	Yes	Yes
OVA (VMware)	Full	Yes	Yes	Yes	Yes	Yes
OVF + VMDK	Full	Yes	Yes	Yes	Yes	Yes
VHD/VHDX (Hyper-V)	Full	Yes	Yes	Yes	Yes	Yes
RAW disk image	Full	Yes	Yes	Yes	Yes	Yes
vSphere export	Full	Yes	Yes	Yes	Yes	Yes

Windows VM Offline Workflow

VirtIO Driver Injection (No Internet)

- Pre-download virtio-win.iso on connected network
- Contains: viostor, vioscsi, NetKVM, balloon, qxlodod
- h2kvmctl mounts guest disk, injects drivers into Windows registry
- Creates boot-critical driver entries for VirtIO SCSI/block
- No Windows Update or online activation required
- Supports: Windows Server 2012R2 through 2025, Win 10/11

Linux VM Offline Workflow

- initramfs rebuild uses only locally installed kernel modules
- `dracut --force --add-drivers "virtio_blk virtio_scsi virtio_net"`
- fstab: convert device names to UUID or LVM paths
- GRUB: update boot parameters for virtio console
- Network: regenerate interface configs for virtio-net
- No dnf/apt calls — pure local operations

Security & Compliance for Air-Gap

FIPS 140-2, STIG hardening, and audit controls for classified migration operations.

Compliance Framework Mapping

Framework	Requirement	hyper2kvm Implementation	Evidence / Artifact
FIPS 140-2	Validated cryptographic modules	Uses OpenSSL FIPS provider; AES-256-GCM for encryption	FIPS mode: <code>fips-mode-setup --check</code> returns enabled
DISA STIG	Hardened OS configuration	Runs on STIG-compliant RHEL 9; no additional ports opened	STIG Viewer checklist export, SCAP scan results
NIST 800-53	Access control, audit logging	All operations logged to syslog; requires root or libvirt group	Audit log: <code>/var/log/hyper2kvm/migration.log</code>
CNSSI 1253	Classified system controls	No network egress; all operations local; media encryption	Network capture shows zero outbound connections
ICD 503	IC information systems	Two-person integrity for media handling; GPG-signed manifests	Signed manifest + DD Form 1149 chain-of-custody
HIPAA	PHI protection during transfer	Encrypted at rest + in transit; audit trail; access controls	BAA-compatible; encryption verification in manifest

Security Controls

Two-Person Integrity (TPI)

- Media export requires two authorized personnel
- Split-knowledge passphrase: each person holds half
- Manifest signed by both exporter and verifier
- Chain-of-custody log with timestamps and signatures
- Media sealed in tamper-evident bags for transport
- Import verification by two different personnel

Audit Trail Without SIEM

- Local syslog: all h2kvmctl operations logged
- JSON-structured logs for post-migration analysis
- Log rotation with integrity checksums (SHA-256)
- Export logs to removable media for offline SIEM ingest
- Timestamp source: local hardware clock or GPS receiver
- Log fields: user, action, source, target, checksum, status

USB Device Policy Enforcement

Write-Blocker Integration

- USBGuard: whitelist only approved transfer devices
- `usbguard allow-device <device-id>`
- Block all USB storage by default on target hosts
- Approved devices: IronKey D500S, Apricorn Aegis only
- Audit: all USB insert/remove events logged
- Auto-lock: device auto-locks after transfer completes

- Forensic write-blockers for source media verification
- Tableau T356789iu or CRU WiebeTech models
- Ensures source VMware disks are not modified during export
- Required for: evidence preservation, classified data handling
- `h2kvmctl` reads source in read-only mode by default
- Verification: compare source checksums pre/post export

Deployment Architecture Patterns

Four proven patterns for air-gapped KVM/KubeVirt deployment.

Pattern 1: Single-Host Portable Lab

- **Use case:** Field deployments, mobile command, SCIF
- Hardware: ruggedized server (Dell R660xs or HPE DL360)
- OS: Rocky 9 STIG-hardened, FIPS enabled
- Storage: 2x NVMe in RAID 1, LUKS encrypted
- VMs: 5-15 per host, all local storage
- Migration: USB transfer, single h2kvmctl run
- Recovery: full host image on encrypted USB backup
- **Cost: \$8K hardware + \$0 software**

Pattern 2: Multi-Host Sneakernet Sync

- **Use case:** Air-gapped data center, 50-200 VMs
- 3-5 hosts with shared NFS or local LVM storage
- Sneakernet sync: export VM changes to USB weekly
- Pacemaker HA cluster (no network to external)
- Local DNS/DHCP/NTP within air-gapped network
- Migration waves: 10-20 VMs per transfer cycle
- Dedicated migration host processes imports
- **Cost: \$40K hardware + \$13K RHEL**

Pattern 3: Air-Gapped K3s + KubeVirt

- **Use case:** Modern workloads, containers + VMs
- K3s lightweight Kubernetes (single binary, 50MB)
- KubeVirt for VM workloads alongside containers
- Longhorn for replicated storage (no Ceph complexity)
- Harbor registry for air-gapped container images
- All images pre-loaded via USB transfer
- GitOps with local Gitea (no GitHub/GitLab cloud)
- **Cost: \$50K hardware + \$0 software**

Pattern 4: Cross-Domain with Data Diode

- **Use case:** UNCLASS → SECRET transfer, IC networks
- Data diode: Owl DualDiode or Waterfall Unidirectional
- Export side: h2kvmctl streams to diode sender
- Receive side: h2kvmctl reassembles from diode receiver
- One-way: physically impossible to reverse data flow
- Automated: scheduled transfers without manual USB
- Bandwidth: 1-10 Gbps depending on diode model
- **Cost: \$80K-\$200K diode + infrastructure**

Recommended: Pattern 2 for Most Air-Gapped Environments

Multi-host with sneakernet sync provides the best balance of capability, cost, and security. Start with Pattern 1 for POC, graduate to Pattern 2 for production. Pattern 3 for organizations with container strategy. Pattern 4 only when continuous cross-domain transfer is mandated.

Implementation Checklist & Timeline

Phase-by-phase plan for 50-VM air-gapped migration.

50-VM Air-Gap Migration Timeline

Phase	Activity	Duration	Personnel	Key Tasks	Deliverables
1	Preparation (Connected)	1 week	2 engineers	Build offline repos, download VirtIO drivers, bundle container images, prepare encrypted transfer media	Offline bundle (~4 GB), encrypted USB drives, chain-of-custody forms
2	Source Export	3 days	2 engineers (TPI)	Export VMs from VMware, create manifests, encrypt, verify checksums, seal media	50 VM images on encrypted media, signed manifests
3	Physical Transfer	1 day	Courier + 2	Chain-of-custody handoff, secure transport, receiving verification	Verified media at destination, custody log
4	Target Setup	2 days	2 engineers	Install OS from offline repos, configure KVM/libvirt, setup networking, STIG harden	Production-ready KVM hosts, SCAP scan reports
5	Conversion	3 days	1 engineer	Run h2kvmctl for all 50 VMs, fix initramfs/fstab, inject VirtIO, generate XML	50 converted QCOW2 images + libvirt domain XMLs
6	Validation	2 days	2 engineers	Boot test each VM, verify networking, application smoke tests, performance baseline	Test results, performance benchmarks, sign-off
7	Cutover	1 day	3 engineers	Shutdown VMware VMs, final delta sync if applicable, go-live on KVM	Production VMs running on KVM, VMware powered off

Total elapsed: ~2 weeks | Total effort: ~80 person-hours

Cost Comparison: Air-Gapped VMware vs KVM

3-Year Air-Gapped VMware (50 VMs)

- Year 1 licensing: \$176,185
- Year 2 licensing: \$176,185 (no discount for air-gap)
- Year 3 licensing: \$176,185

3-Year Air-Gapped KVM (50 VMs)

- hyper2kvm migration (one-time): \$0
- Migration labor (80 hours): \$12,000
- RHEL subscriptions (optional): \$13,000/yr x 3 = \$39,000

- Offline patch media: \$15,000/yr x 3 = \$45,000
- VMware admin (air-gap specialist): \$140,000/yr x 3
- **3-Year Total: \$993,555**

- OS patches from ISO (quarterly): \$0
- Linux admin: \$120,000/yr x 3 = \$360,000
- **3-Year Total: \$411,000**

3-Year Savings: \$582,555 (59% reduction)

Air-gapped environments benefit most from open-source KVM — no license activation, no phone-home telemetry, no per-VM fees, no annual audit surprises. hyper2kvm's fully offline workflow eliminates the #1 barrier to migration in disconnected environments.